

Block-Level Urban Growth Prediction from Google Open Buildings Temporal Data

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Abstract—We use the Google Open Buildings 2.5D Temporal dataset (Sirko et al., 2023) to predict where new construction will occur in 25 cities across Africa, South Asia, South-East Asia and Latin America. Rather than predict at the individual pixel level, we group pixels into $17\text{ m} \times 17\text{ m}$ blocks, which reduces sensor noise while keeping building-level detail. Five neural models are compared against a simple baseline on 500 location patches drawn at random from five cities not seen during training. The best model – a Block U-Net trained on seven years of building history – finds real new construction 2.4 times more often than a random guess of the same number of blocks. Adding road proximity and city-centre distance as extra inputs brings no improvement, and using only the most recent year instead of all seven makes almost no difference.

I. INTRODUCTION

Knowing in advance where new buildings will appear helps city planners allocate roads, schools, water supply and other infrastructure before demand outpaces capacity. In much of the Global South, official land-use records are incomplete or outdated, so satellite-derived building maps are often the only available source of this information. Sirko et al. (2023) publish yearly per-pixel building presence maps at roughly 4 m resolution from 2016 onward, covering most of the developing world.

In this work we use seven consecutive years of those maps (2016–2022) to predict where new buildings will appear in 2023. The model assigns a score to every small area in a city; areas with higher scores are more likely to see new construction. We evaluate how useful these scores are by checking how much more often the highest-scored areas actually see new construction compared to randomly chosen areas of the same total size.

We run a controlled experiment comparing five neural model variants against a simple baseline. The experiment is designed to answer three questions: (i) does producing a spatially detailed output (a score per 17 m block) improve on producing just one summary score per neighbourhood? (ii) does adding a decoder that reconstructs spatial detail help? (iii) does using all seven years of history improve on using only the most recent year? We also test whether adding road proximity and city-centre distance as extra inputs helps when the seven-year history is already available. Both variables are motivated by Brandily and Rauch (2024), who show that road density causally predicts population growth in Sub-Saharan African

cities, and by Wang et al. (2025), who use distance-to-roads and distance-to-administrative-centres as standard predictors in a continent-scale study of African urban expansion.

II. DATA

The Google Open Buildings 2.5D Temporal dataset (Sirko et al., 2023) provides one map per year from 2016 to 2023, where each pixel holds a value between 0 and 1 representing how confident the model is that the pixel is covered by a building that year. The maps are produced by a machine-learning pipeline that learns from high-resolution commercial satellite imagery and then runs on freely available Sentinel-2 imagery (Sirko et al., 2023), which is why it covers the whole developing world.

A. City selection and splits

Twenty-five cities are included, spread across Africa, South Asia, South-East Asia and Latin America to cover a wide range of urban growth patterns and climates. For each city we read a roughly $0.04^\circ \times 0.04^\circ$ window (about 4.5×4.5 km) for each year and resize it to a 1024×1024 pixel grid. Cities are divided into three groups (Table I): 15 for training the models, 5 for selecting the best model, and 5 held back entirely for final evaluation. No data from the test cities is used at any point during training.

TABLE I
CITY SPLITS.

Split	Cities
Train	new_cairo, accra, nairobi, lagos, addis_ababa, dhaka, delhi, karachi, chennai, jakarta, ho_chi_minh, lima, bogota, quito, medellin
Val	manila, phnom_penh, kampala, kigali, bangkok
Test	mexico_city, sao_paulo, dar_es_salaam, lahore, hanoi

B. Training patches

From each city’s map we extract 128×128 pixel patches (roughly 552×552 m on the ground). To ensure the model sees enough examples of actual construction, 70% of patches are required to contain at least some new buildings in 2022→2023, and 30% are areas that already have buildings but no significant recent growth. This deliberate imbalance is only applied

to the training and validation data. For evaluation we use a separate set of 500 patches drawn uniformly at random from the five test cities, with no growth requirement, as described in Section IV. In total, 2,522 patches are used: 1,542 for training, 480 for validation, and 500 for the unbiased test.

C. From pixel labels to 17 m blocks

The raw per-pixel labels have a noise problem. When we looked at every pixel labelled as a new building in the 2022→2023 transition, we found that 76% of them were completely isolated from neighbouring new-building pixels. A real building covers tens of square metres; a single 4 m pixel is far too small to be a real structure. These isolated pixels are most likely errors or detection artefacts in the source model, not actual new construction.

To reduce this noise, we group every 4×4 pixel cell into one $17 \text{ m} \times 17 \text{ m}$ block. The block label is the fraction of its 16 pixels that are marked as new buildings. A typical building footprint spans 1–3 blocks, so this grouping still captures real construction events but largely averages out the single-pixel errors. The average fraction of a block that becomes newly built during training is 3.16%.

D. Auxiliary input channels

Two extra data sources are added to test whether they improve the best model:

- **Road proximity.** For each city we download the drivable road network via OSMnx (Boeing, 2017) and compute, for every pixel, how far it is from the nearest road. We convert this distance d into a score $1/(1 + d/300 \text{ m})$ that is close to 1 next to a road and decreases smoothly with distance. The motivation is that Brandily and Rauch (2024) find, using data from over 1,800 African cities, that road density in a city centre causally predicts how fast that city grows.
- **Distance from city centre.** A simple score for each pixel indicating how far it is from the geographic centre of the city window, normalised so the centre is 0 and the corner is 1. Wang et al. (2025) use an equivalent variable as a key spatial predictor in a simulation of African urban expansion.

III. METHODS

A. What the model predicts

Let B_t be the building presence map for year t . A pixel is counted as a new building if it had no building in year t and had a building in year $t + 1$, defined as:

$$g(i, j) = \mathbf{1}[B_t(i, j) < 0.5] \cdot \mathbf{1}[B_{t+1}(i, j) \geq 0.5], \quad (1)$$

where $\mathbf{1}[\cdot]$ equals 1 when the condition is true and 0 otherwise. In words: the model counts a building as new if the probability of a building was below 50% in one year and 50% or above the next year. The block-level target is simply the fraction of the 16 pixels in the block for which this is true. All models output a score between 0 and 1 for each block.

We train with Huber loss (Huber, 1964) with $\delta = 0.02$. Huber loss works like squared error for small mistakes, but for larger mistakes it switches to a gentler penalty. This matters here because almost all blocks have a score of exactly 0 (no growth), and a handful have small positive values. Using standard squared error would make the rare large errors dominate training and destabilise it; Huber loss avoids this.

B. Models

Six configurations are compared (Table II).

TABLE II
MODELS COMPARED. ALL BLOCK-LEVEL MODELS OUTPUT A SCORE FOR EACH 17 M BLOCK IN THE PATCH. “PER PATCH” MEANS A SINGLE NUMBER IS PREDICTED FOR THE ENTIRE 552 M NEIGHBOURHOOD.

Model	Output	Params
Mean baseline	per patch	N/A
Patch CNN	per patch	166k
Block CNN	per block	289k
Block U-Net (2022 only)	per block	1.57M
Block U-Net (7 yrs)	per block	1.57M
Block U-Net + roads/centre	per block	1.57M

Mean baseline. Always predicts the average growth rate seen during training. This is the simplest possible reference point.

Patch CNN. A convolutional neural network that reads the entire 552 m patch and outputs a single growth-rate number for the whole neighbourhood. That same number is then assigned to every block inside the patch. This model can tell us roughly how much a neighbourhood will grow, but it has no way to say *where* within the neighbourhood growth will concentrate.

Block CNN. The same type of convolutional network, but the architecture is set up so that the output is a grid of scores, one per 17 m block, rather than a single number. The network progressively compresses the input image down to a small summary, then reads off one score per output block. Crucially, it has no mechanism to recover spatial detail that was lost during compression.

Block U-Net (2022 only). This model follows the U-Net design introduced by Ronneberger et al. (2015). Like the Block CNN it compresses the image step by step, but it also has a second half that expands back to the original resolution. At each expansion step, the model receives a direct shortcut connection from the corresponding compression step, which lets it recover fine-grained spatial detail when predicting which specific blocks will grow. The input here is only the 2022 building presence map – one snapshot in time. This lets us measure how much the most recent year alone is worth.

Block U-Net (7 yrs). The same U-Net architecture, but the input is seven building presence maps stacked together (2016 through 2022) – one map per year. Comparing this to the 2022-only version shows how much historical context adds.

Block U-Net + roads/centre. Same as the 7-year U-Net but with two extra inputs: road proximity and distance from city centre, described in Section II-D. This tests whether

knowing about roads and location adds anything when the model already sees all seven years of building maps.

C. Training

All neural models are trained using the Adam optimiser (Kingma and Ba, 2015), which is a standard gradient-based method for learning model weights. We use a learning rate of 10^{-3} , batch size 16, and stop training early when performance on the validation set stops improving (after 7 consecutive epochs with no gain). Seed 42 is fixed throughout; saved model files mean no experiment needs to be re-run.

IV. EVALUATION

A. Unbiased test set

The training data was deliberately collected to include many examples of construction, which inflates how often positive examples appear. For the final evaluation we build a separate, unbiased test set: 500 patches drawn uniformly at random from the five held-out test cities (100 per city), with no requirement for any growth. This mimics real deployment, where the model is applied across an entire city including areas with no new construction. On this test set, 27.1% of blocks have at least one new building in 2022→2023.

B. Metrics

PR-AUC (primary). When the model outputs a score for each block, the scores can be ranked from highest to lowest. PR-AUC summarises how well that ranking works: does the model place the actually-growing blocks near the top of the list? A perfect model scores 1.0; a completely random model scores equal to the fraction of positive blocks (here, 0.271). Davis and Goadrich (2006) show that this metric is more reliable than the commonly used ROC-AUC when positives are rare, which is the case here.

Lift. PR-AUC divided by the fraction of positive blocks (0.271). Lift = 1.0 means the model’s ranking is no better than random. Lift = 2.4 means that if a planner inspects the blocks the model ranked highest, they will find real new construction 2.4 times as often as they would by inspecting a random set of blocks of the same size. We use lift as the headline metric because it has a direct practical interpretation and removes the distortion caused by the 27.1% positive rate.

R^2 , MAE, F1. R^2 measures how much of the variation in actual block growth rates the model’s predictions explain; 0 means the model is no better than always predicting the mean, 1 means perfect. MAE is the average prediction error in percentage points. F1 is the standard precision-recall balance score at the threshold where a block is called positive if at least one of its 16 pixels grew.

V. RESULTS

A. Main comparison

Table III shows the full results. Figure 1 shows the precision–recall curves.

The mean baseline achieves a lift of 1.0 as expected – it predicts every block the same way, so the ranking is

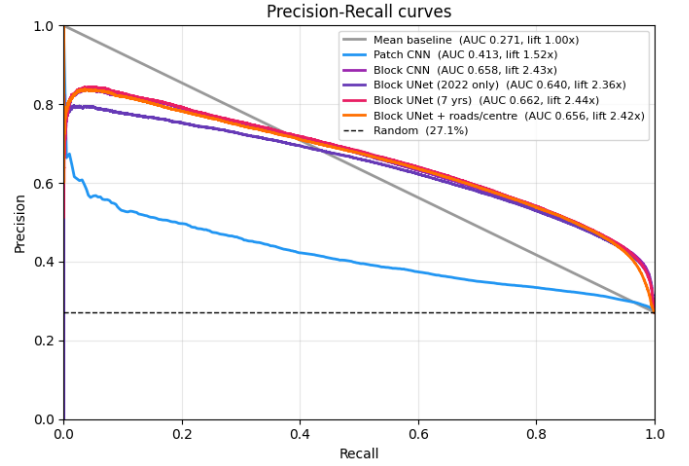


Fig. 1. Precision–recall curves for all models on the unbiased test set. The dashed line shows the score of a random ranking.

meaningless. The Patch CNN raises this to lift 1.52, showing that the model has learned something useful from the seven-year building maps, even though it can only predict a single number per neighbourhood.

The biggest jump in the table is from Patch CNN (lift 1.52) to Block CNN (lift 2.43). These two models use the same underlying network, but the Block CNN outputs a score *per block* rather than one number for the whole neighbourhood. This single change almost doubles the lift, and confirms that the most important capability is knowing *where within* a neighbourhood growth will occur, not just how much growth to expect overall.

The three remaining block-level models – Block CNN, both U-Net variants, and the roads/centre version – all produce very similar results (lift 2.36–2.44, PR-AUC 0.640–0.662). Looking at the precision-recall curves in Figure 1, the curves for these three models largely overlap. The 7-year Block U-Net is technically the best-performing model and is the configuration we recommend, but the differences between the block-level models are small.

B. Does seven years of history help?

Comparing the 1-year Block U-Net (PR-AUC 0.640, lift 2.36) to the 7-year Block U-Net (PR-AUC 0.662, lift 2.44) shows that adding six more years of building history does improve the results slightly, but the gain is modest. The 1-year model already captures about 86% of the improvement over random. Hafner et al. (2024) offer a relevant explanation: in their study of urban change detection from satellite image series, they found that historical images only provide large gains when the model has a dedicated mechanism – such as an attention module – that explicitly compares information across different time steps. When years are simply treated as separate input channels, as we do here, the model has to figure out the time ordering on its own, which limits how much it can benefit.

TABLE III
PERFORMANCE ON THE 500-PATCH UNBIASED TEST SET (2022→2023, 5 HELD-OUT CITIES). POSITIVE BLOCK RATE IS 27.1%; LIFT = PR-AUC / 0.271. BOLD MARKS THE BEST VALUE PER COLUMN.

Model	PR-AUC	Lift	R ²	MAE %pt	F1
Mean baseline	0.271	1.00	−0.002	5.00	0.000
Patch CNN	0.413	1.52	0.033	5.02	0.302
Block CNN	0.658	2.43	0.250	3.28	0.499
Block U-Net (2022 only)	0.640	2.36	0.200	3.32	0.501
Block U-Net (7 yrs)	0.662	2.44	0.253	3.25	0.526
Block U-Net + roads/centre	0.656	2.42	0.227	3.21	0.450

C. Do roads and city-centre distance help?

The +roads/centre model achieves essentially the same results as the 7-year Block U-Net (lift 2.42 vs. 2.44, PR-AUC 0.656 vs. 0.662). The extra inputs add nothing. Our interpretation is that the seven years of building maps already contain this information implicitly: roads show up as consistent linear gaps in the building maps across all years, and the location of the city centre is visible from the density of the built-up area. The model appears to extract this information from the maps directly, making the explicit road and distance channels redundant.

This is surprising given that both variables are well-supported in the literature. Brandily and Rauch (2024) show they causally predict city growth, and Wang et al. (2025) use them as standard predictors. However, those studies operate without temporal building maps; here the seven-year history acts as a substitute.

D. Why do all block-level models plateau?

All block-level models converge around lift 2.4. We attribute this ceiling to noise in the labels themselves. Even after grouping into 17 m blocks, a block with just one or two noisy pixels still gets a small positive label. Improving beyond this point would likely require cleaner ground truth, for example by requiring a new building to be detected across multiple years rather than in a single transition.

E. Visualisation of predictions

Figure 2 shows block-level predictions for the patch with the most actual growth in each test city. Each coloured cell is one 17 m block. The green overlay marks where buildings actually appeared. The orange–red overlay shows the model’s predicted score, with brighter red meaning the model was more confident. The Patch CNN panel shows a uniform colour across the whole patch – it assigns the same score to every block. The Block U-Net panel shows varying colours, with higher scores concentrated in the areas that actually grew.

VI. CONCLUSION

A Block U-Net trained on seven years of building presence maps achieves a lift of 2.44 on a 500-patch test set drawn at random from five held-out cities – meaning its top-scored blocks contain real new construction 2.4 times as often as a random selection. The most important finding is that producing

a score per 17 m block rather than a single score per neighbourhood is by far the biggest contributor to this result. Adding road proximity, city-centre distance, or additional years of history makes little difference. All block-level models perform similarly, and the ceiling appears to be set by noise in the underlying labels rather than model capacity.

REFERENCES

- G. Boeing, “OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks,” *Computers, Environment and Urban Systems*, vol. 65, pp. 126–139, 2017.
- P. Brandily and F. Rauch, “Within-city roads and urban growth,” *Journal of Regional Science*, vol. 64, no. 4, pp. 1236–1264, 2024.
- J. Davis and M. Goadrich, “The relationship between precision-recall and ROC curves,” in *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, 2006, pp. 233–240.
- S. Hafner, Y. Ban, and A. Nascetti, “Continuous urban change detection from satellite image time series with temporal feature refinement and multi-task integration,” *arXiv preprint arXiv:2406.17458*, 2024.
- P. J. Huber, “Robust estimation of a location parameter,” *The Annals of Mathematical Statistics*, vol. 35, no. 1, pp. 73–101, 1964.
- D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *International Conference on Learning Representations (ICLR)*, 2015.
- O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.
- W. Sirko, E. A. Brempong, J. T. C. Marcos, A. Annkah, A. Korme, M. A. Hassen, K. Sapkota, T. Shekel, A. Diack, S. Nevo, J. Hickey, and J. Quinn, “High-resolution building and road detection from Sentinel-2,” *arXiv preprint arXiv:2310.11622*, 2023.
- J. Wang et al., “Analysis of the spatial pattern of urban expansion in African countries under different SSP scenarios,” *Land*, vol. 14, no. 3, art. 558, 2025.

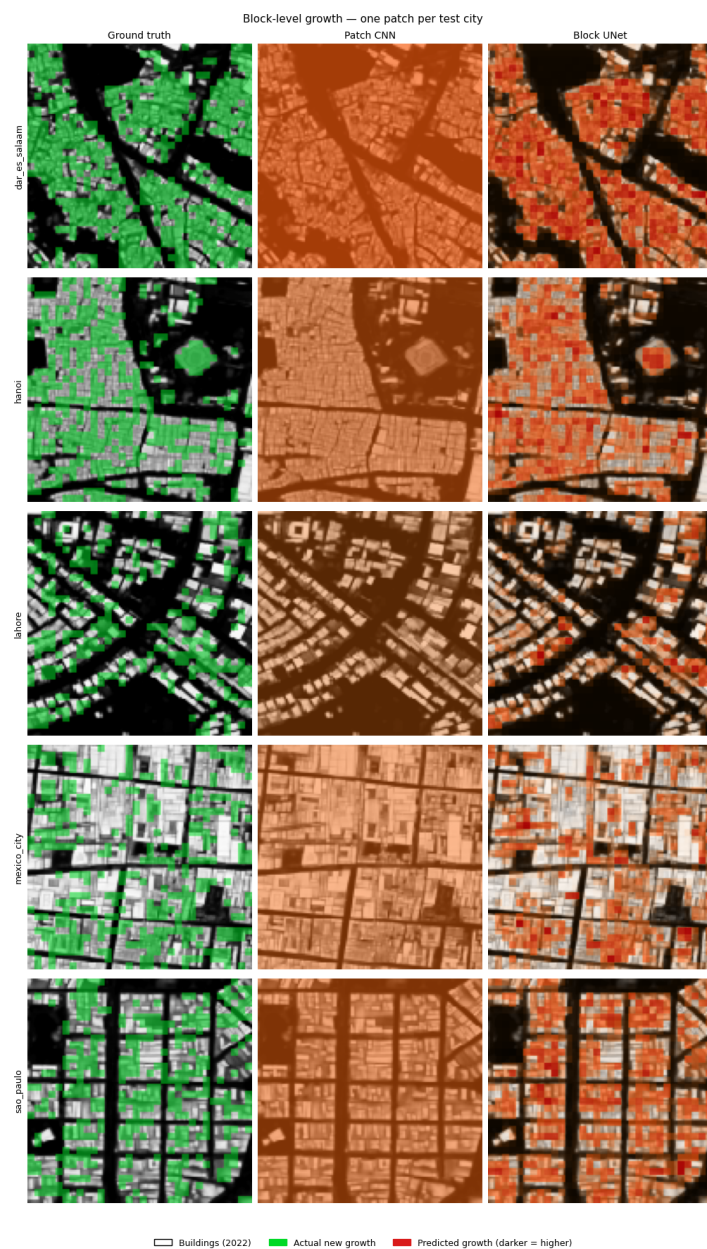


Fig. 2. Block-level predictions for one patch per test city. Left: actual new construction (green). Middle: Patch CNN predictions. Right: Block U-Net predictions. Brighter orange-red means a higher predicted growth score.